

Cascade Game Rules

COMP30024 Artificial Intelligence

February, 2026

In *Cascade*, towers rise and fall like empires. Build towering stacks to unleash devastating waves across the board, pushing your opponent's forces toward oblivion. But beware: what rises together can fall apart, and the edge of the board is always closer than it seems.

Overview

Cascade is a two-player strategy game played on an 8x8 grid. Each player controls a set of tokens that can be stacked to form *stacks* of varying heights. After an initial setup phase, players take turns performing one of three actions: **MOVE**, **EAT** or **CASCADE**. The objective is to eliminate all of the opponent's tokens by strategically building and toppling stacks.

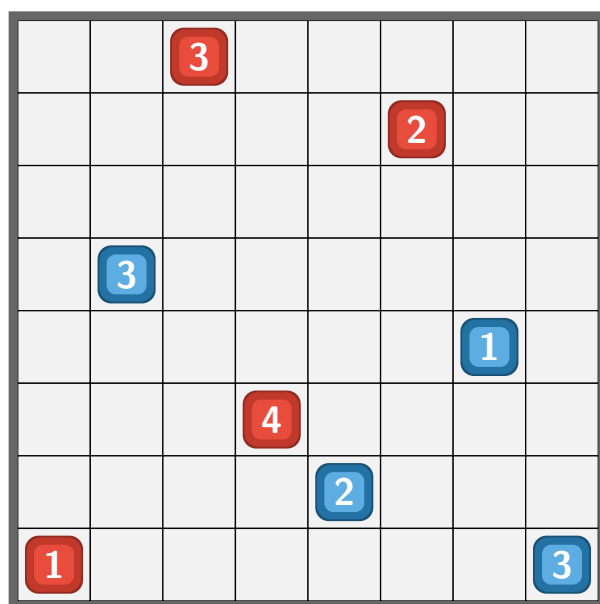


Figure 1: An example game of *Cascade*.

Game Board

We use a two-dimensional coordinate system to describe the positions of cells on the game board (Figure 2). Formally, a valid board *coordinate* is an integer pair (r, c) , $0 \leq r \leq 7$, $0 \leq c \leq 7$, where r is the *row* on the board and c is the *column*.

A given coordinate may be directly *adjacent* to up to **four** other coordinates in the cardinal directions. For example, $(1, 2)$ is adjacent to four coordinates:

- $(1, 3)$ – RIGHT
- $(1, 1)$ – LEFT
- $(0, 2)$ – UP
- $(2, 2)$ – DOWN

Those on the edges or corners of the board naturally have fewer adjacent coordinates. For example, the coordinate $(7, 7)$ only has two: $(6, 7)$ (UP) and $(7, 6)$ (LEFT). The board does not wrap around, so $(7, 7)$ is not adjacent to $(0, 7)$ or $(7, 0)$.

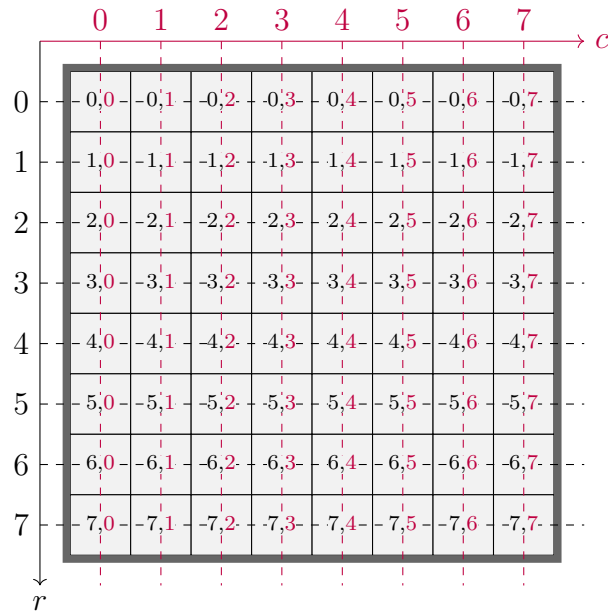


Figure 2: The coordinate system used in Cascade.

Gameplay

A game of **Cascade** proceeds in two phases:

1. The **placement phase**: players take turns placing four stacks, each of height 3, on an initially empty board.
2. The **play phase**: players take turns performing one of three actions — **MOVE**, **EAT** or **CASCADE** — to outmanoeuvre and eliminate the opponent's tokens.

By convention, **Red** always takes the first turn. Throughout both phases, players alternate turns.

PLACE (Placement Phase)

The game begins with an empty board. During the placement phase, players take a total of 8 turns (four each). On each turn, a player plays a **PLACE** action to create a new stack of height 3 on any empty cell.

After the very first placement of the game, there is one restriction: a stack may not be placed on a cell that is *adjacent* to any of the opponent's existing stacks.

Once all 8 placements have been made, each player has 12 tokens on the board (four stacks of height 3), and the play phase begins.

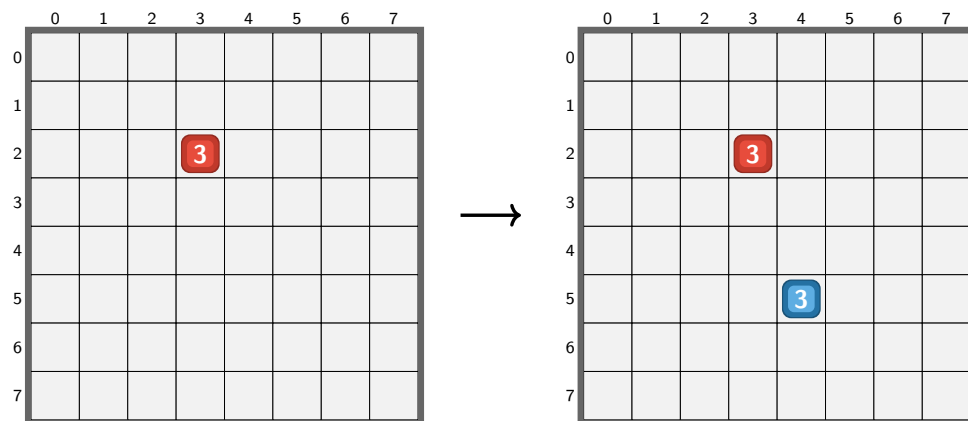


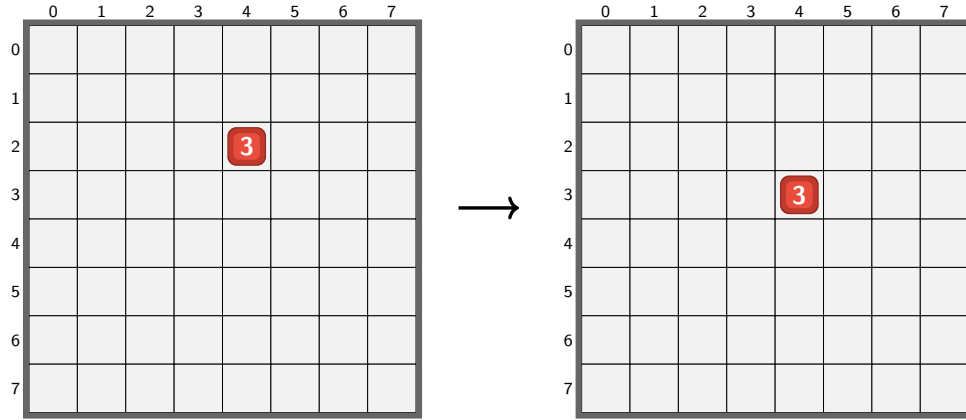
Figure 3: **Blue** plays **PLACE** at (5, 4), creating a new stack of height 3. Note that **Blue** cannot place adjacent to **Red**'s existing stack at (2, 3).

MOVE

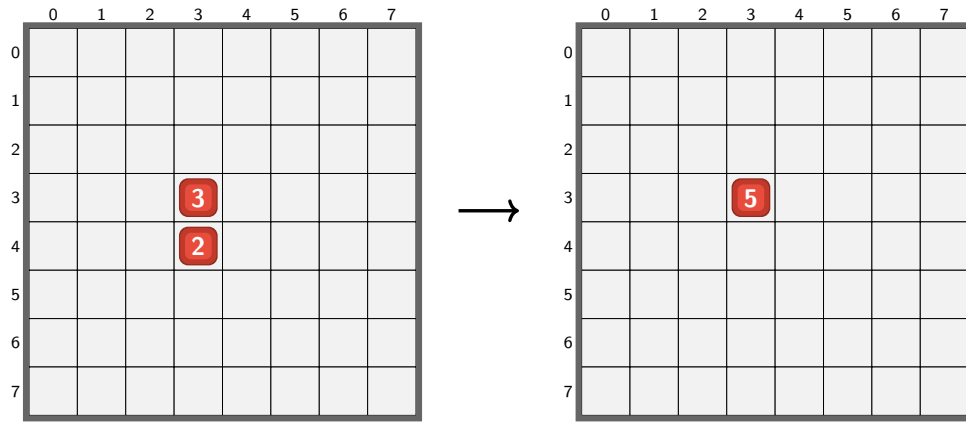
A player may play a **MOVE** action to move one of their stacks to an adjacent cell in a cardinal direction. There are two possible outcomes depending on the destination cell:

- **Relocate:** If the destination cell is *empty*, the entire stack moves there. The source cell becomes empty.
- **Merge:** If the destination contains a *friendly* stack (same colour), the two stacks merge into one. The resulting stack has a height equal to the sum of both heights.

A MOVE action cannot target a cell containing an enemy stack — use EAT for that instead.



(a) Red plays $\text{MOVE}[(2, 4), \text{DOWN}]$, relocating to an empty cell.



(b) Red plays $\text{MOVE}[(4, 3), \text{UP}]$, merging with a friendly stack. The resulting height is $2 + 3 = 5$.

Figure 4: Two examples of the MOVE action: (a) relocating to an empty cell and (b) merging with a friendly stack.

EAT

A player may play an EAT action to capture an adjacent enemy stack in a cardinal direction. The attacking stack must have a height *greater than or equal to* the target stack's height. The attacker moves to the target's cell, replacing the enemy stack. The captured tokens are removed from the game entirely, and the source cell becomes empty.

The attacker's height does not change after eating — the captured tokens are eliminated, not absorbed.

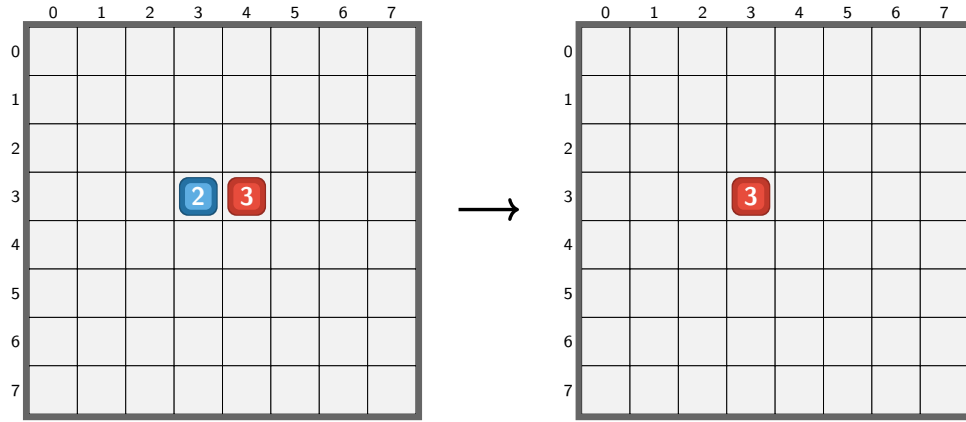


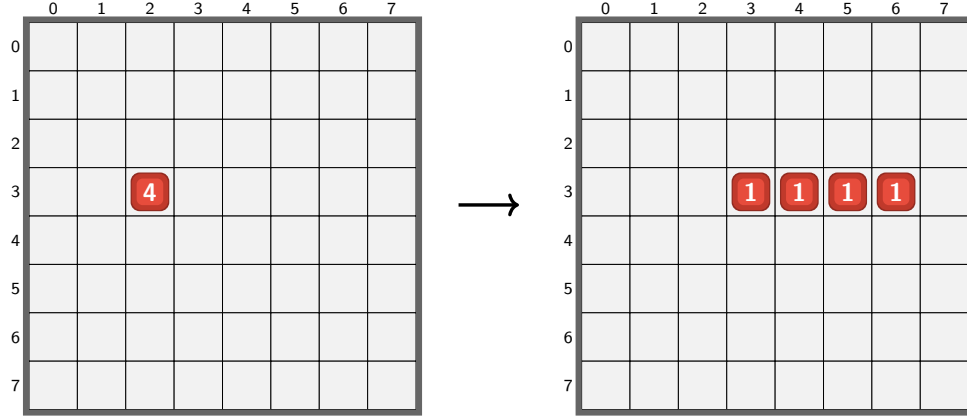
Figure 5: **Red** plays **EAT** (3,4) **LEFT**, capturing **Blue**'s stack. The attacker's height (3) $\geq$ the target's height (2), so the action is valid. **Blue**'s 2 tokens are removed from the game. (Note that the attacker's height remains 3, as the captured tokens are not absorbed.)

CASCADE

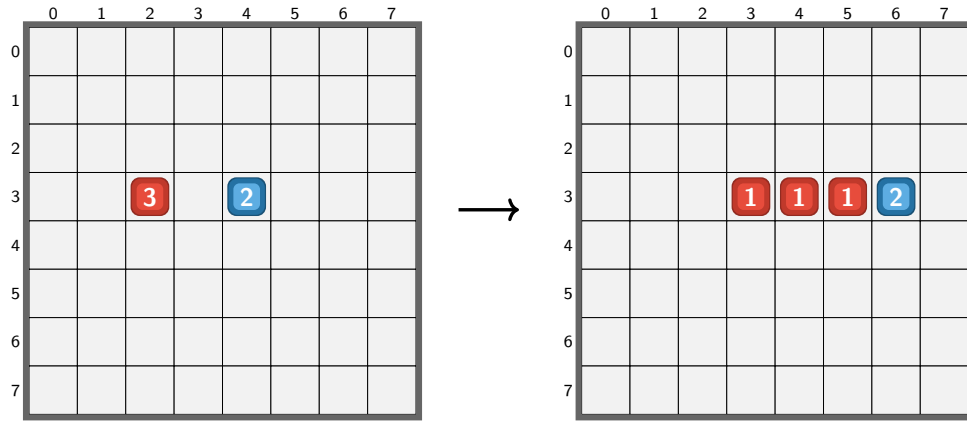
A player may play a **CASCADE** action to spread one of their stacks across multiple cells in a cardinal direction. Only stacks with height 2 or greater can cascade.

When a stack of height h cascades in a given direction:

1. The original stack is removed from its cell.
2. For each position from 1 to h cells away in the cascade direction, a single token of the cascading player's colour is placed.
3. If a cell along the path contains an existing stack (of either colour), that stack is *pushed* (not merged) one cell further in the cascade direction before the new token is placed. Pushed stacks can in turn push other stacks ahead of them.
4. Any stack that gets pushed off the edge of the board is *eliminated* from the game.
5. Any token from the cascading stack that would land off the edge of the board is also eliminated.



(a) Red plays CASCADE (3,2) RIGHT. The stack of height 4 spreads into four individual tokens.



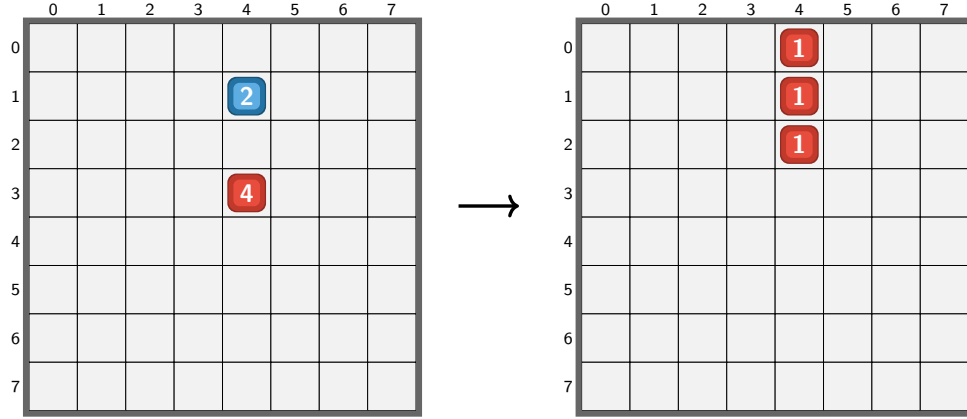
(b) Red plays CASCADE (3,2) RIGHT. The second token pushes Blue's stack from (3,4) to (3,5), and the third token pushes it again to (3,6).

Figure 6: Two examples of the CASCADE action: (a) spreading into empty space and (b) pushing an opponent's stack along the cascade direction.

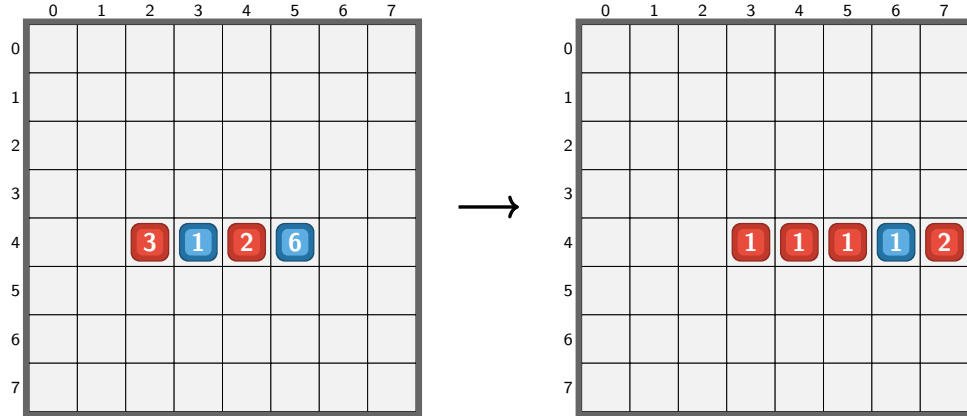
Ending the Game

A game of Cascade can end in a **win** for either player, or a **draw**. There are three possible end conditions:

1. **Elimination.** If all of a player's tokens are removed from the board, their opponent wins immediately.
2. **Turn limit.** If 300 turns have been played during the play phase (not counting the 8 placement turns) without a winner, the game ends. The player with more total tokens on the board is declared the winner. If both players have the same number of tokens, the game is a draw.
3. **Threefold repetition.** If the same board position occurs three times (with the same player to move), the game immediately ends in a draw, regardless of either player's material advantage.



(a) Red plays CASCADe (3,4) UP. Blue's stack at (1,4) is pushed off the top of the board as well as 1 of Red's tokens.



(b) Red plays CASCADe (4,2) RIGHT. Each cascading token pushes the stacks ahead of it. Importantly, the friendly stack (height 2) is *pushed* not merged. It retains its original height at its new position (4,7). The enemy stack (height 1) is also pushed, not eaten, because the cascade action is not an attack.

Figure 7: Two more examples of the CASCADe action: (c) pushing a stack off the edge of the board, with a token also lost over the edge, and (d) pushing both friendly and enemy stacks without merging.